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FINAL PROJECT REPORT

**Natural Language Processing
Sentiment Analysis Platform**

Sentiment Analysis Service for Multilingual Review Intelligence

Course: NLP501 – AI in Production: From Models to Systems

Topic: NLP – Sentiment Analysis Service

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multilingual_sentiment

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Abstract

This project presents a complete natural language processing platform for sentiment analysis, aspect-based sentiment analysis, sarcasm detection, explainability, data versioning, model training, and production deployment. The system was implemented as a full-stack engineering project using a web frontend, a REST backend, container-based deployment, Prometheus/Grafana observability, and MLflow/DVC for reproducible machine-learning workflows.

The core inference service supports multiple execution modes. In baseline mode the platform uses a pretrained transformer for direct sentiment inference; in fine-tuned mode it uses a multilingual transformer with PEFT/LoRA adapters for sentiment and sarcasm tasks; and in ONNX modes it serves exported FP32 or INT8 models through ONNX Runtime. Aspect-based sentiment analysis is implemented through a zero-shot classifier, while token-level explainability is provided by SHAP. Language support is enforced at the backend level through a lightweight local detector that accepts English and Vietnamese.

The project addresses the operational problem of converting unstructured customer feedback into structured, machine-readable sentiment signals that can support review monitoring, customer-service triage, and product analysis. The implementation includes a reproducible data pipeline for dataset ingestion, preprocessing, validation, training preparation, evaluation, and ONNX export; experiment tracking for metrics and artifacts; request and inference monitoring; and automated checks for code quality and test execution.

Evaluation artifacts in the repository show a baseline test-set result of 0.8135 accuracy and 0.7389 macro F1 on 799 SemEval test samples, together with a fine-tuned evaluation artifact reporting 0.7926 overall F1 on a sampled evaluation run. Data-quality validation reports a clean processed dataset of 3,831 samples with zero null-ratio violations and one distribution warning for the missing location aspect category. The repository also contains ONNX benchmark artifacts, fairness reports, SHAP visualization outputs, and deployment configuration for a production-style multi-service stack.

The main contribution of the project is not a single model checkpoint but an end-to-end sentiment analysis system with reproducible engineering controls, model explainability, monitoring, and deployment artifacts. The platform also documents explicit limitations, including the absence of authenticated access control, the mock nature of the batch-status workflow, and the fact that document/audio upload capabilities described in frontend notes are not implemented in the backend.

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1 Chapter 1 Introduction

1.1 Background

Sentiment analysis is the task of identifying the emotional or evaluative orientation expressed in text. In modern NLP systems it is commonly applied to customer reviews, social posts, survey responses, and support tickets. The repository implements this problem in a restaurant-review domain using transformer-based text classification, aspect-based sentiment analysis, sarcasm detection, and post-hoc explainability.

The practical value of sentiment analysis comes from its ability to transform unstructured feedback into structured evidence. The platform is designed for customer-feedback interpretation, where the same input can be evaluated at the sentence level and at the aspect level. The design combines overall sentiment, ABSA, and SHAP explanations into a single response.

1.2 Problem Statement

Manual sentiment inspection does not scale to large feedback streams and introduces inconsistency across reviewers. It is also difficult to use manually generated sentiment labels in downstream systems because the output is not machine-readable, not reproducible, and not directly measurable. The project addresses this by implementing an API-based pipeline that produces deterministic model outputs, machine-readable response schemas, and monitoring signals suitable for production deployment.

1.3 Project Objectives

Table 1: Project objectives by layer

Objective type	Implemented objective
Functional objectives	Single prediction, batch CSV prediction, SHAP explanation, health check, and evaluation trigger endpoints.
Technical objectives	Typed API contracts, DVC-managed data flow, MLflow experiment tracking, ONNX export, Prometheus metrics, and Docker deployment.
AI objectives	Multilingual sentiment inference, sarcasm detection via a second adapter, aspect-based sentiment extraction, and token-level explainability.
Deployment objectives	Multi-service container deployment using Docker Compose with frontend, backend, monitoring, and tracking services.

1.4 Scope

Table 2: Scope definition

In Scope	Out of Scope
Single and batch inference, ABSA, SHAP explainability, language detection, DVC data processing, LoRA fine-tuning, ONNX export, metrics, dashboards, alerting, and test automation.	User authentication, rate limiting, online learning, active learning, real-time message queues, document upload, audio upload, and fully asynchronous batch-job orchestration.

1.5 Contributions

- Built a backend service in FastAPI with typed request and response schemas.
- Implemented a configurable inference engine with baseline, finetuned, and ONNX execution modes.
- Added ABSA through zero-shot classification and added sarcasm detection through a dedicated adapter path.
- Implemented SHAP-based explanation generation and frontend explanation rendering.
- Built a DVC pipeline for raw data extraction, preprocessing, validation, training preparation, evaluation, and export.
- Added MLflow logging for evaluation and training runs.
- Added Prometheus metrics, alert rules, and Grafana provisioning.
- Built an Angular frontend with chat interaction, batch CSV upload, theme switching, and explainability display.

2 Chapter 2 Literature Review and Theoretical Foundation

2.1 Sentiment Analysis

Sentiment analysis studies how opinions, preferences, emotions, and evaluations can be inferred from text. In practice, the task has evolved from lexicon-based polarity scoring to supervised neural classification and, more recently, to transformer-based language understanding. The project uses this classical sentiment classification problem as the primary interface for customer-feedback intelligence.

The main challenge in sentiment analysis is that opinion polarity is often context-dependent. Negation, contrastive conjunctions, intensifiers, domain-specific vocabulary, and sarcasm can all alter the surface meaning of a sentence. For that reason, the project does not rely on a single label only; it combines sentence-level sentiment, aspect-level sentiment, sarcasm detection, and explainability to reduce ambiguity in interpretation.

2.2 Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis decomposes a review into semantically meaningful targets such as food, service, ambiance, price, or location, and then estimates the polarity associated with each target. This is more useful than a single document-level label when the same review contains mixed opinions. A customer may praise the food while criticizing the service, and a document-level classifier would otherwise collapse that distinction.

The project adopts a pragmatic ABSA design based on zero-shot inference. This choice avoids the need for a separate annotated ABSA corpus for every domain-specific aspect while still enabling structured output. The trade-off is that zero-shot ABSA is more sensitive to prompt phrasing and may be less stable than a domain-trained aspect classifier.

Table 3: Sentiment analysis research concepts and project usage

Concept	Role in theory	Role in this project
Document sentiment	Estimates overall polarity of a text.	Forms the primary prediction returned by the platform.
Aspect sentiment	Associates polarity with a specific target.	Produces structured feedback for food, service, and related aspects.
Sarcasm/irony	Handles polarity reversal and figurative language.	Provides an auxiliary binary signal for difficult samples.
Explainability	Makes prediction evidence visible.	Returns SHAP token contributions for user inspection.

2.3 Transformer Architecture

Transformer models address sequence modeling through tokenization, embedding, attention, and stacked encoder blocks. Tokenization converts raw text into subword units, which helps the model handle rare words and multilingual text. Embeddings map these tokens into dense vectors, while self-attention computes token-to-token interactions so the model can represent context.

For classification, the project uses a transformer encoder followed by a classification head. The head maps the final hidden representation to class probabilities. This design is well suited to sentiment analysis because it captures both local cues and longer-range dependencies without requiring recurrent computation.

2.4 Multilingual NLP

Multilingual NLP extends the same modeling framework across languages. The main challenges are script variation, tokenization differences, vocabulary imbalance, and uneven data availability. The project handles multilingual input by using a multilingual backbone and a lightweight language gate that accepts English and Vietnamese inputs. This design is narrow but deliberate: it avoids overclaiming broad multilingual coverage while still supporting the datasets actually used in the repository.

2.5 PEFT and LoRA

Parameter-efficient fine-tuning modifies only a small subset of model parameters, which lowers compute cost and storage requirements. LoRA is the mechanism used in this project: low-rank matrices are injected into selected attention projections, allowing the backbone to remain mostly frozen while task-specific behavior is learned in compact adapter weights.

The primary advantage of LoRA is that it reduces training cost and makes it practical to maintain separate adapters for different tasks, such as sentiment and sarcasm detection. The main trade-off is that task adaptation capacity is constrained by the low-rank approximation, so performance can lag behind a full fine-tune if enough data and compute are available.

2.6 Explainable AI

Explainable AI is important in sentiment systems because a prediction is often only useful if users can understand why it was made. SHAP is used in this project because it provides a consistent additive explanation framework grounded in Shapley value

theory. Each token receives a contribution score that reflects how much it supports or suppresses the predicted class relative to a baseline.

The explanation interface is intended for human review rather than formal audit. It helps a lecturer, examiner, or evaluator understand whether the model is reacting to sentiment-bearing words, negation, or aspect-specific terms. The repository does not include a dedicated explanation-fidelity benchmark, so the explainability layer is validated qualitatively through visual inspection.

2.7 MLOps Concepts

The system reflects several MLOps concepts: DVC is used to make data and pipeline stages reproducible, MLflow is used to track metrics and artifacts, monitoring is used to observe runtime behavior, and containerization is used to preserve execution consistency. These tools provide the operational bridge between model development and deployable software.

Table 4: Theoretical concepts and engineering implications

Concept		Engineering implication
Transformer self-attention		Better handling of context and long-distance dependency.
PEFT / LoRA		Lower training cost and easier adapter management.
SHAP explainability		User-facing interpretability and token-level evidence.
MLOps toolchain		Reproducible datasets, experiments, and deployment artifacts.

3 Chapter 3 Requirements Analysis

3.1 Stakeholders

Table 5: Stakeholder analysis

Stakeholder	Role in the system
Users	Submit text, inspect sentiment, view aspect-level results, and request explanations through the frontend or HTTP API.
Administrators	Operate the deployment stack, monitor uptime, and inspect metrics and dashboards.
Developers	Extend the backend, frontend, pipeline, tests, and deployment configuration.
Researchers	Reproduce experiments, review metrics, and inspect generated artifacts such as fairness reports and SHAP plots.

3.2 Functional Requirements

Table 6: Functional requirement matrix

ID	Priority	Requirement	Implementation evidence
FR-01	Must	Predict sentiment for a single text input.	The platform returns an overall sentiment label and confidence score for a single submitted message.
FR-02	Must	Return aspect-level sentiment alongside the overall label.	The platform identifies supported aspects and returns per-aspect sentiment for each relevant category.
FR-03	Must	Provide SHAP-based explanations.	The platform exposes token-level explanation values for a submitted prediction.
FR-04	Must	Support CSV batch prediction.	The platform accepts a CSV file, validates the expected text column, and returns row-level inference results.
FR-05	Must	Expose health information.	The platform reports model availability and supported language coverage.
FR-06	Should	Detect language and reject unsupported languages.	The platform detects English and Vietnamese inputs and rejects unsupported languages.
FR-07	Must	Expose Prometheus metrics.	The platform exports operational metrics for scraping.
FR-08	Must	Evaluate the model and log artifacts.	The platform can run evaluation workflows and persist metrics artifacts.
FR-09	Should	Provide a usable browser frontend.	A browser-based interface supports chat-style interaction and batch analysis.
FR-10	Should	Support monitoring and alerting.	The deployment includes scraping and alerting configuration for operational visibility.

3.3 Non-Functional Requirements

Table 7: Non-functional requirements

Category	Requirement	Evidence
Performance	Keep inference usable on CPU and support ONNX acceleration.	The system provides a CPU-oriented runtime path and exported ONNX models for lower-latency deployment options.
Reliability	Return well-typed responses and structured error messages.	Typed request and response contracts plus structured exception handling enforce consistent API behavior.
Scalability	Keep the service stateless and containerized.	Docker Compose deploys separate services with a shared network.
Maintainability	Use explicit contracts and typed Python modules.	The codebase separates API contracts, data processing, inference, training, and deployment concerns.
Security	Avoid runtime dependency downloads where possible.	The runtime image ships with required artifacts and avoids unnecessary external fetches.
Usability	Provide a chat-style frontend and batch upload modal.	The browser interface uses a conversational workflow with batch-upload support.
Availability	Provide live-ness/health endpoints and monitoring.	Health checks, metrics exposure, scraping, and dashboards support uptime monitoring.

3.4 Use Case Analysis

Table 8: Use cases and actors

Actor	Use cases
User	Submit a text, read sentiment result, inspect ABSA output, open an explanation panel, upload a CSV file.
Administrator	Monitor request metrics, inspect Grafana dashboards, review alerts, restart services.
Developer	Modify pipeline stages, retrain adapters, export ONNX graphs, run tests, inspect MLflow runs.
Researcher	Compare baseline and fine-tuned metrics, inspect fairness reports, review SHAP plots.

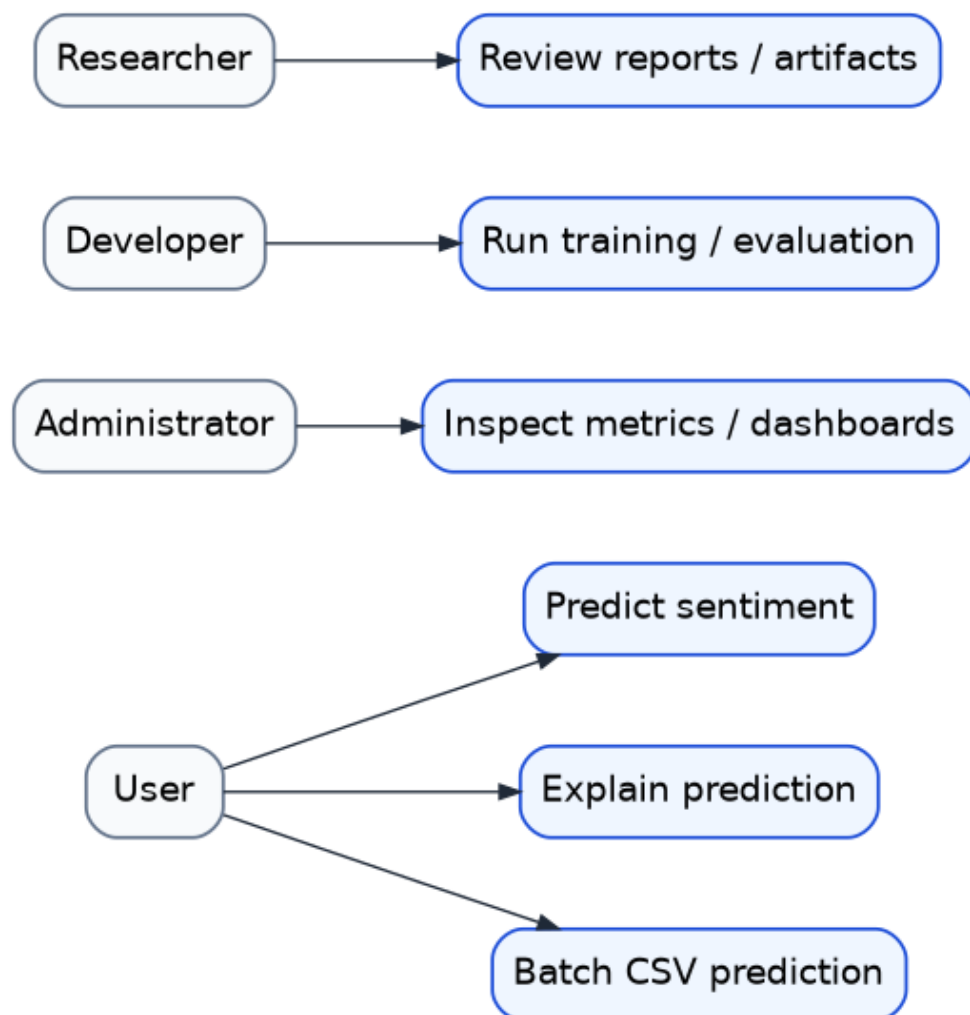


Figure 1: Use case diagram for the platform

4 Chapter 4 System Architecture

4.1 Overall Architecture

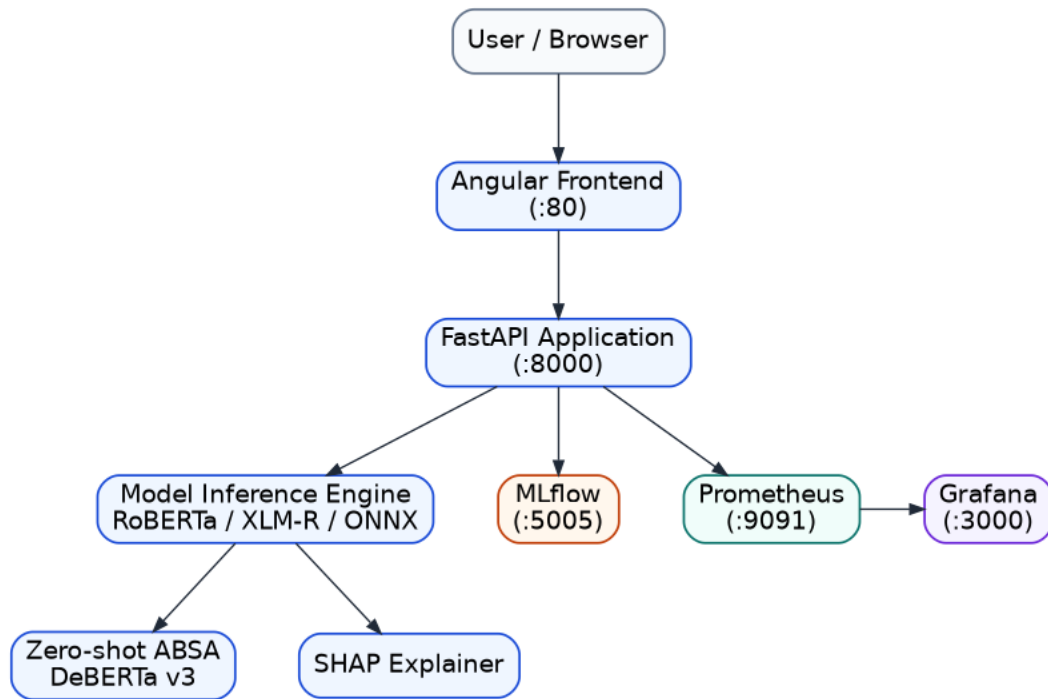


Figure 2: Overall system architecture

4.2 Context Diagram

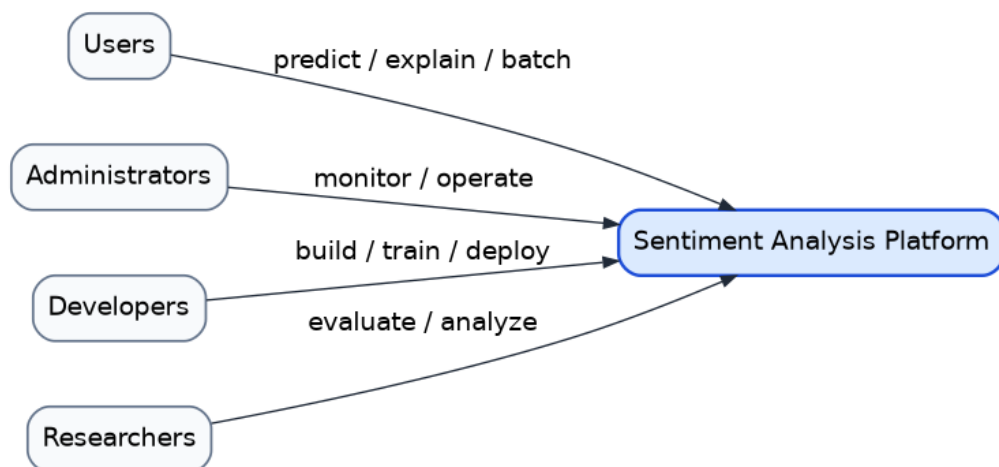


Figure 3: Context diagram

4.3 Component Diagram

Table 9: Core components

Component	Subsystem	Responsibility
API layer	Backend service	Request validation, route dispatch, exception mapping, metrics exposure.
Inference engine	Model runtime	Baseline, fine-tuned, and ONNX inference; ABSA; sarcasm flagging; SHAP explanations.
Data pipeline	Data processing layer	Dataset ingestion, preprocessing, validation, and report generation.
Training	Training subsystem	LoRA adapter training, class balancing, evaluation utilities.
Frontend	Web application	Chat UI, batch upload modal, explanation view, theme switching.
Monitoring	Observability layer	Request and inference metrics plus alerting rules.

4.4 Architectural Design Principles

The architecture is organized around five principles. Modularity is achieved by separating the API, inference engine, data pipeline, training workflow, and observability stack. Reproducibility is addressed through parameterized data and experiment workflows. Scalability is supported by containerized services that can be scaled independently. Maintainability is improved through typed contracts and bounded responsibilities. Observability is built into the runtime through metrics, dashboards, and evaluation artifacts.

Table 10: Architectural principles and implementation outcomes

Principle	Implementation outcome
Modularity	Separate services handle API, inference, training, and monitoring responsibilities.
Reproducibility	Data and experiment workflows are parameter-driven and artifact-based.
Scalability	The system can be deployed as independent containers with shared network contracts.
Maintainability	The interfaces between subsystems are explicit and typed.
Observability	The runtime emits metrics, logs, evaluation artifacts, and dashboard inputs.

4.5 Design Decisions

The system was shaped by a set of explicit engineering trade-offs. FastAPI was preferred over Flask because the project benefits from typed request validation and automatic API documentation with minimal boilerplate. Angular was preferred over React because the team-oriented front-end architecture already followed a component-and-service pattern with reactive state and opinionated structure. LoRA was preferred over full fine-tuning because it reduces training cost and makes it practical to maintain multiple task adapters. ONNX was preferred over native PyTorch serving because it provides a deployment-oriented runtime path and supports quantization. SHAP was preferred over LIME because the project needed token-level additive explanations that are easier to align with transformer outputs.

Table 11: Major technology decisions

Decision	Chosen option	Alternative	Trade-off
API frame- work	FastAPI	Flask	Better typing and validation at the cost of a slightly more opinionated stack.
Frontend framework	Angular	React	Stronger architectural conventions at the cost of a steeper initial learning curve.
Fine-tuning method	LoRA / PEFT	Full fine-tuning	Lower cost and smaller adapters at the expense of reduced capacity.
Serving for- mat	ONNX Runtime	Native PyTorch	Better deployment portability at the expense of export complexity.
Explainability	SHAP	LIME	Stronger additive interpretation at the cost of higher computation.

4.6 Deployment Diagram

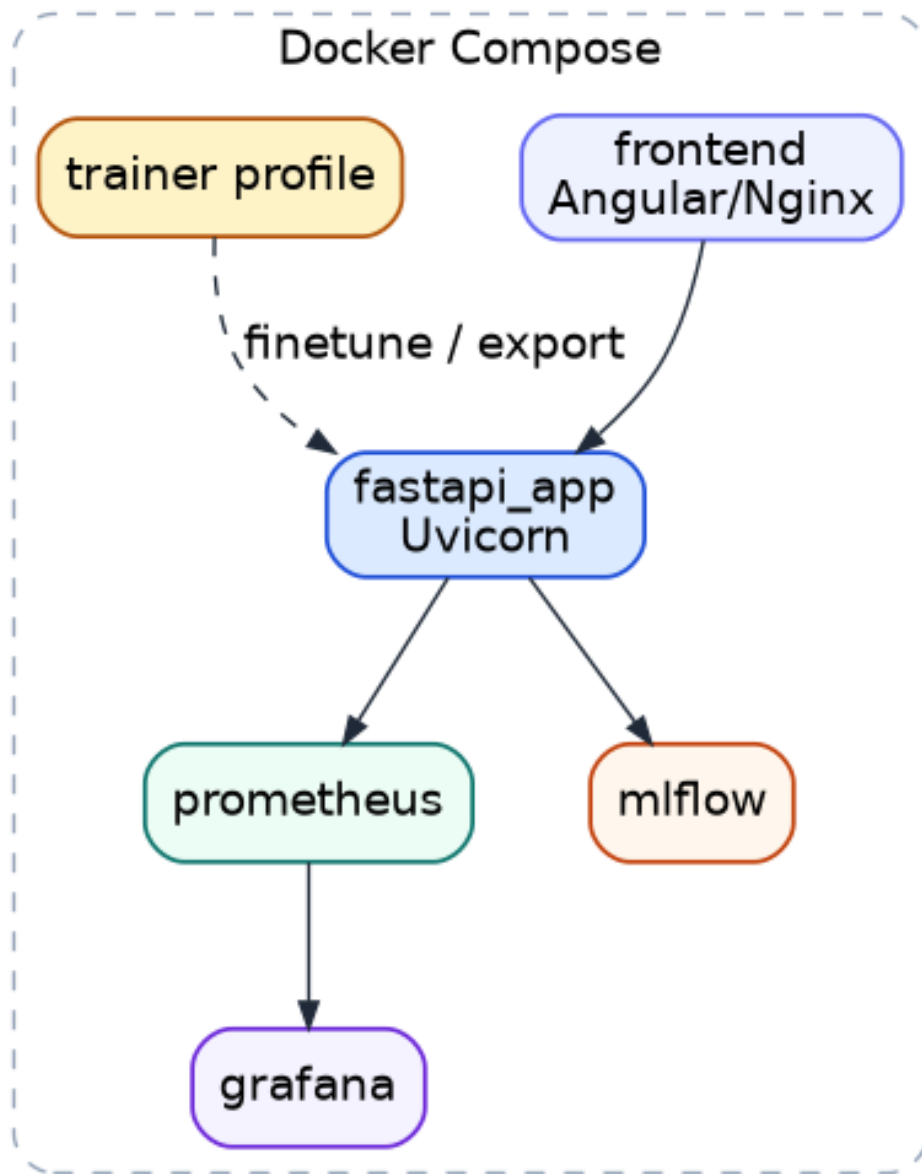


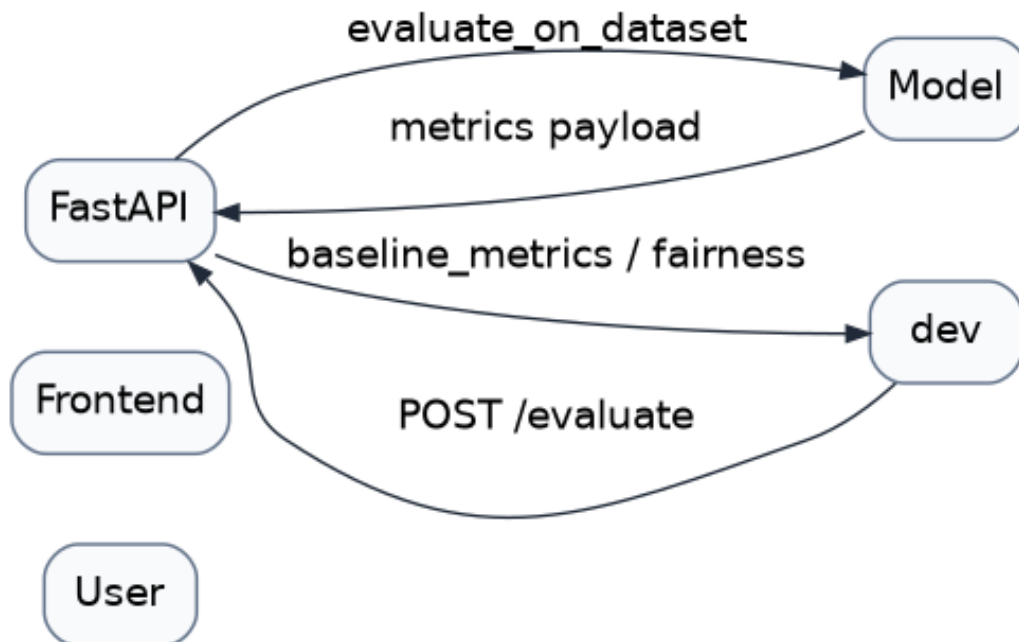
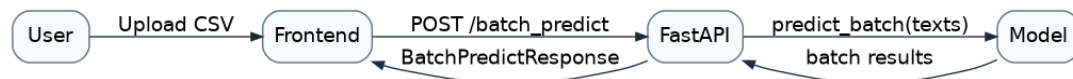
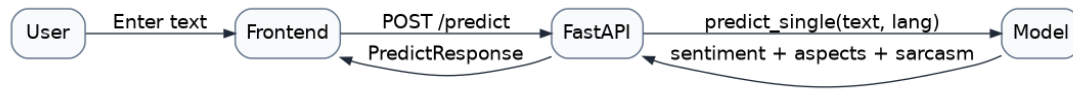
Figure 4: Deployment diagram

4.7 Data Flow Diagram



Figure 5: Data flow diagram

4.8 Sequence Diagrams



5 Chapter 5 Technology Stack

Table 12: Technology stack and usage

Layer	Technology	Purpose	Usage in this project
Frontend	Angular, RxJS, Nginx	Browser UI and static hosting	The web application implements a chat-style interface with reactive state and API proxying.
Backend	FastAPI, Pydantic, Uvicorn	REST API and validation	The backend exposes health, prediction, explanation, batch, evaluation, and metrics endpoints.
AI / ML	Transformers, PEFT, LoRA, ONNX Runtime, SHAP, zero-shot ABSA	Model inference, tuning, export, explainability	The ML stack supports baseline, fine-tuned, and optimized runtime inference modes.
Data engineering	DVC, Pandas, scikit-learn, PyYAML	Reproducible pipeline and data transforms	The pipeline supports parameterized ingestion, preprocessing, validation, and reporting.
Infrastructure	Docker, Docker Compose, Prometheus, Grafana	Container deployment and observability	The deployment stack includes container orchestration, scraping, and dashboarding.
Experiment tracking	MLflow	Metrics, params, artifacts	Experiment history and model artifacts are logged for reproducibility.
Testing	pytest, pytest-cov, httpx	Verification	The test suite covers contracts, data transforms, model helpers, API, and performance-related behavior.

5.1 Purpose, advantages, and usage

The frontend stack is appropriate for a chat-like user experience because it enables reactive state handling and lightweight deployment behind Nginx. The backend stack is appropriate for this service because FastAPI integrates typed schemas, automatic OpenAPI generation, and straightforward ASGI middleware for observability. The machine-learning stack is appropriate because transformer-based classification, PEFT/LoRA adapters, ONNX export, and SHAP are directly supported by the Python ecosystem used in the repository.

6 Chapter 6 Data Engineering Pipeline

6.1 Dataset Sources

Table 13: Data sources used by the repository

Source	Format	Role in the project
SemEval-2014 Restaurant Reviews	XML to CSV	Baseline sentiment and aspect labels extracted from restaurant-review annotations.
tweet_eval_irony	Hugging dataset	Face Sarcasm/irony training and evaluation.
multilingual_sentiment	Hugging dataset	Face English and Vietnamese sentiment training.
Generated reports	JSON / PNG	Data-quality, class-balance, metrics, fairness, and SHAP artifacts.

6.2 Dataset Profiling

The data pipeline profiles each dataset before training. Profiling focuses on sample count, label coverage, null ratios, and text-length distributions. This step is important because the project combines multiple datasets with different label spaces and different class frequencies. Without profiling, silent distribution shifts could propagate into training and evaluation.

Table 14: Dataset profiling summary

Dataset	Samples	Primary labels	Profiling note
SemEval reviews	3,831	positive, negative, neutral	Highly imbalanced toward positive labels.
Irony corpus	Balanced subset	irony, non-irony	Suitable for adapter-based binary classification.
Multilingual sentiment	Mixed-language split	positive, negative, neutral	Supports English and Vietnamese evaluation.

6.3 Data Collection

The data acquisition layer directly parses SemEval XML files when available and also supports Hugging Face-backed dataset retrieval for sarcasm and multilingual sentiment tasks. The pipeline standardizes fields such as text, label, language, split, and source. The raw data schema is validated before downstream processing.

6.4 Data Distribution Analysis

The processed sentiment distribution is skewed toward positive examples, while the sarcasm dataset is substantially more balanced. This difference matters because imbalance affects both loss optimization and evaluation stability. The project therefore treats sentiment and sarcasm as separate training tasks with different balancing strategies rather than forcing them into one unified sampling policy.

6.5 Data Cleaning

The preprocessing pipeline is implemented as an ordered transform chain:

Table 15: Preprocessing transforms

Transform	Behavior
LabelMapper	Removes <code>conflict</code> aspect labels and remaps categories such as <code>ambience</code> to <code>ambiance</code> and <code>anecdotes/miscellaneous</code> to <code>general</code> .
SentimentDeriver	Derives sentence-level sentiment from aspect labels using the <code>negative_priority</code> strategy.
TextCleaner	Lowercases, normalizes whitespace, and removes empty strings.
DuplicateRemover	Deduplicates by <code>text</code> while keeping the first occurrence.
LengthFilter	Keeps text lengths between 3 and 2,000 characters.
Splitter	Stratified validation split using seed 42 and a 0.1 validation ratio.

6.6 Data Validation

The validator checks schema presence, minimum sample counts, null ratios, expected label sets, and distribution warnings. The recorded quality report contains 3,831 total samples, zero null-ratio violations, and a warning that location is missing from the aspect distribution. That report is evidence that the data-quality layer is functioning as intended.

6.7 Data Validation Rules

Table 16: Validation rules

Rule		Purpose
Required fields present		Prevents malformed records from entering training or evaluation.
Minimum sample threshold		Avoids training on degenerate or incomplete datasets.
Null-ratio limit		Detects silent data corruption or incomplete extraction.
Label-set validation		Ensures labels match the expected task taxonomy.
Distribution warning		Surfaces underrepresented labels and missing aspect categories.

6.8 Dataset Splitting

The processed SemEval dataset was split into train, validation, and test partitions. The quality report shows:

Table 17: Processed dataset summary

Split	Samples	Null ratio	Observed labels
Train	2,728	0.0	positive 1,528; negative 669; neutral 531
Validation	304	0.0	positive 170; negative 75; neutral 59
Test	799	0.0	positive 501; negative 184; neutral 114

6.9 Data Quality Assurance

The class-balance report shows that SemEval sentiment is severely imbalanced, with an imbalance ratio of 11.48x on the combined split inventory. This evidence justifies the class-weighting and oversampling logic used in the finetuning pipeline. By contrast, the sarcasm dataset is close to balanced, which is why the sarcasm task uses class weights without minority oversampling.

6.10 Reproducibility Strategy

Reproducibility is addressed by versioning datasets, parameterizing pipeline stages, and storing report artifacts after each pipeline run. Because the same data preparation and training choices are reused across evaluation and export, the repository can reproduce

the same model family without manual intervention. This is more important than the exact choice of a single model checkpoint because the project is intended as a repeatable engineering system.

6.11 Data Lineage

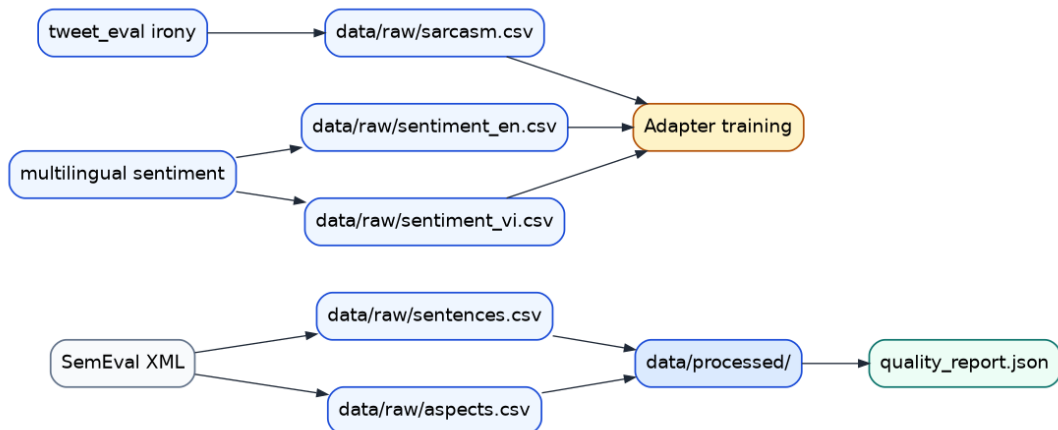


Figure 10: Data lineage diagram

7 Chapter 7 Model Development

7.1 Baseline Model

The baseline model is the direct inference configuration used for single-text sentiment analysis. In baseline mode, the backend loads a pretrained sentiment transformer. This mode is used for direct sentiment inference, ABSA, and SHAP explanations.

7.2 Problem Formulation

The model development problem can be stated as a multi-task text understanding problem. Given an input review, the platform must estimate the overall sentiment label, identify the sentiment of relevant aspects, detect sarcasm when present, and provide an explanation that is understandable to a human reviewer. Formally, the system maps an input text sequence to a structured output composed of class probabilities, aspect-level labels, a sarcasm flag, and explanation scores.

Table 18: Model output space

Output element	Meaning
Overall sentiment	Coarse polarity of the full input text.
Aspect sentiment	Polarity associated with a detected aspect target.
Sarcasm flag	Auxiliary signal for language that may invert literal sentiment.
Explanation scores	Token-level attribution values used for interpretability.

7.3 Sentiment Classification Theory

Sentiment classification is a supervised learning task in which text is mapped to discrete classes. The project uses the standard three-way polarity formulation: negative, neutral, and positive. This formulation is appropriate for review data because it handles both clear opinions and neutral statements without forcing all inputs into a binary positive/negative split.

The main difficulty is that sentiment is rarely expressed through a single obvious word. The model must infer polarity from context, compositional syntax, negation, and domain-specific expressions. For example, a review may contain both praise and criticism, and the final label must reflect the dominant or most informative sentiment pattern.

7.4 Transformer Pipeline

The transformer inference pipeline follows a consistent sequence: input text is normalized and tokenized into subword units, token embeddings are combined with positional information, self-attention layers contextualize each token with the full sequence, and the final encoder representation feeds a classification head. The classification head outputs probabilities over the sentiment classes and can be shared or adapted across tasks.

Table 19: Transformer pipeline stages

Stage	Role
Input text	Raw review or social text submitted by the user.
Tokenizer	Converts text into model-compatible subword tokens.
Embeddings	Maps tokens into dense vectors for downstream processing.
Encoder layers	Learn contextual dependencies through attention.
Classification head	Produces class probabilities and final labels.

7.5 Fine-tuned Model

The fine-tuned configuration uses a multilingual transformer plus PEFT/LoRA adapters. The repository trains two separate tasks:

- sentiment classification with labels `negative`, `neutral`, `positive`;
- sarcasm detection with labels `non_irony` and `irony`.

The sentiment adapter and sarcasm adapter are loaded together in fine-tuned mode. The sarcasm adapter uses a three-logit head in training so it can share the same backbone, while inference consumes only the first two logits for the binary sarcasm flag.

7.6 LoRA Architecture

LoRA inserts low-rank update matrices into selected projection layers while keeping the backbone largely frozen. This preserves the general linguistic knowledge acquired during pretraining while learning task-specific behavior in a compact parameter space. In this project the approach is especially useful because sentiment and sarcasm adapters can be maintained separately without duplicating the full backbone.

Table 20: LoRA design rationale

Aspect	Rationale
Small rank	Limits the number of trainable parameters and keeps adapters compact.
Targeted layers	Focuses adaptation on attention projections where contextual interactions are strongest.
Separate adapters	Allows different tasks to be learned and deployed independently.
Frozen backbone	Preserves general language knowledge while reducing training cost.

7.7 Sarcasm Detection Model

Sarcasm training uses the irony dataset, three training epochs, a learning rate of $2e-4$, batch size 16, and class weighting. The repository does not implement a separate rule-based sarcasm detector; sarcasm prediction is model-based and exposed through a dedicated response field.

7.8 Aspect-Based Sentiment Analysis

ABSA is implemented with a zero-shot classifier. The model first identifies whether a text contains any of the supported aspects, then performs per-aspect sentiment

classification using the aspect-specific hypothesis template. The threshold for keeping an aspect is 0.45.

7.9 Sarcasm Detection Design

Sarcasm is treated as an auxiliary classification task rather than a rule-based filter. This is important because sarcasm often depends on context and can co-occur with both positive and negative lexical cues. The project therefore trains a dedicated adapter that produces a binary irony signal, which can be attached to the main sentiment prediction as a cautionary indicator.

7.10 Explainability Design

SHAP explainability is implemented in the inference layer. The method predicts the class, creates a SHAP explainer around the active classifier, and returns token strings, token-level SHAP scores, and a base value. The response schema validates that the token list and SHAP value list have equal length.

7.11 Runtime Optimization

Runtime optimization is implemented by exporting fine-tuned adapters to ONNX and quantizing them to an INT8 variant. The purpose of this step is not only raw speed but deployment portability: the exported artifact can be served independently of the training stack and can be compared against the native PyTorch path.

7.12 Quantization Strategy

Quantization reduces numerical precision in exchange for smaller model size and lower memory bandwidth requirements. The project uses post-training quantization to create an INT8 runtime artifact. This is suitable for inference because the model is not being updated during serving, and the deployment target values responsiveness and portability over maximal floating-point fidelity.

7.13 Model Limitations

The model stack remains constrained by its training data and task formulation. ABSA is zero-shot rather than domain-trained, so aspect detection can be sensitive to prompt design. The language gate only accepts English and Vietnamese, which keeps the inference boundary narrow. The sarcasm signal is auxiliary rather than a complete pragmatic interpretation model. These limitations are acceptable for a capstone-scale system but should be addressed before production hardening.

7.14 Training Configuration

Table 21: Training configuration

Item	Configuration
Base model	Multilingual transformer backbone
LoRA rank / alpha / dropout	<code>r=8, lora_alpha=16, lora_dropout=0.05</code>
Target modules	<code>query, value</code>
Classifier saving	Classifier head preserved during adapter export
Sentiment training	5 epochs, learning rate 2e-4, batch size 16, gradient accumulation 2
Sarcasm training	3 epochs, learning rate 2e-4, batch size 16, gradient accumulation 2
Label balancing	Class weights enabled for both tasks; oversampling enabled for sentiment

7.15 ONNX Optimization

The ONNX export pipeline merges each PEFT adapter into the base model, saves a merged intermediate checkpoint, exports to FP32 ONNX, and then quantizes to INT8 using Optimum. Runtime inference is implemented through ONNX Runtime.

7.16 Model architecture

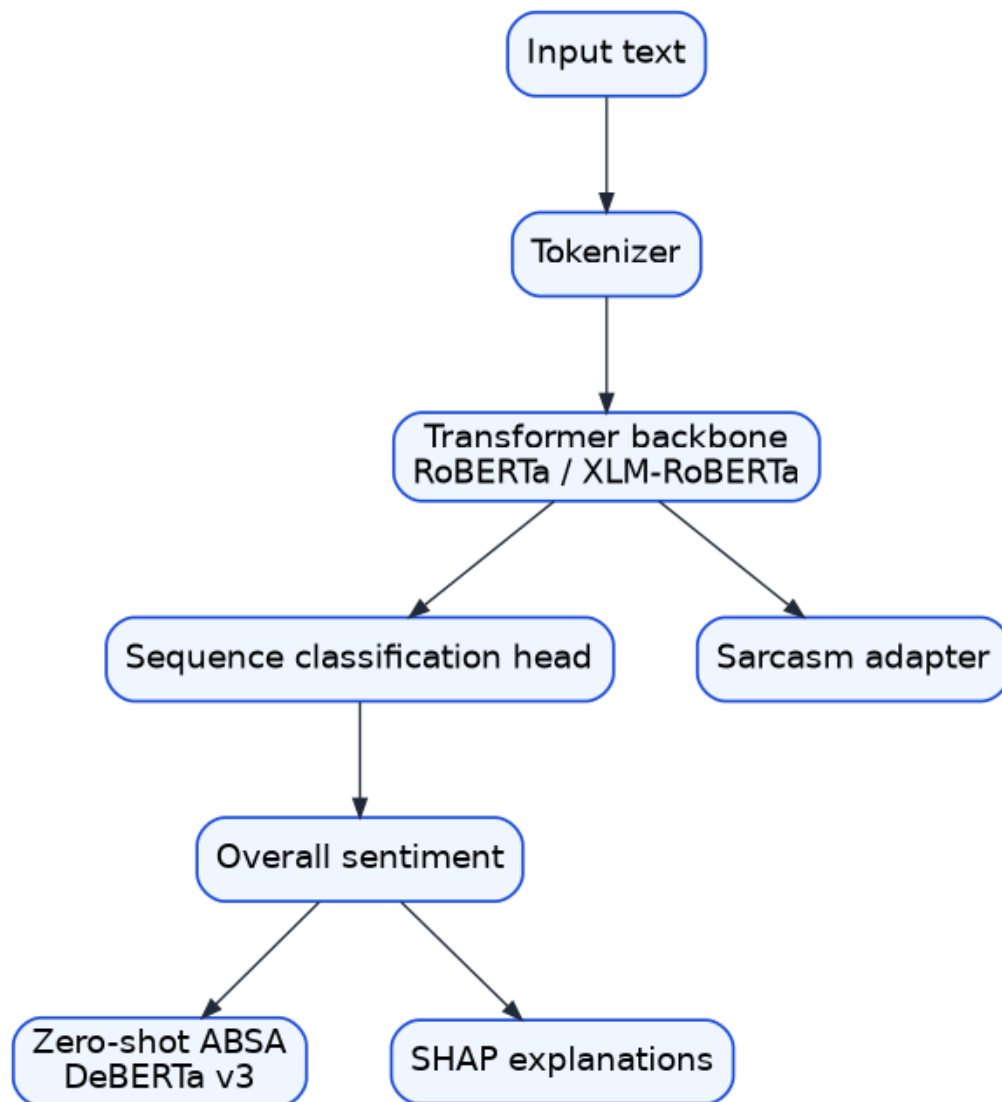


Figure 11: Model architecture

8 Chapter 8 System Implementation

8.1 Frontend Implementation

The frontend is a standalone Angular application with three visible composition units: the root chat window, the message bubble component, and the batch upload modal. State is managed with reactive signals. The root component polls the backend health endpoint and shows an initialization overlay until the backend is ready. The message bubble renders sentiment badges, aspect badges, sarcasm indicators, and a collapsible

SHAP panel. The batch upload workflow accepts a CSV file with a text column, validates the file extension, sends the file to the batch endpoint, and renders result tables and sentiment summaries.

8.2 State Management

Table 22: Frontend state model

State source	Responsibility
Conversation state	Stores the message history and explainability state.
Theme state	Stores user theme preference and persists it in local storage.
Backend readiness state	Controls the initialization overlay and frontend readiness.
Batch upload state	Drives the CSV batch upload workflow.

8.3 UI Workflow

The user enters a message, the service calls the prediction endpoint, the UI renders the overall sentiment and aspect breakdown, and a subsequent explanation button allows the user to request SHAP values. Batch analysis is performed through a dedicated modal. The frontend also includes a language picker and script warning logic; however, the backend only enforces English and Vietnamese support, so unsupported values are not guaranteed to succeed.

8.4 Backend Implementation

The backend is centered around a global model instance created in the application lifespan handler. The application exposes typed routes, custom exception handlers, CORS configuration, monitoring middleware, and a Prometheus metrics endpoint. Validation is performed at the schema layer, while language support is enforced inside the model inference layer.

Table 23: Backend endpoints

Endpoint	Method	Behavior
Health check	GET	Returns model load state, version, and supported languages.
Prediction	POST	Runs language detection, overall sentiment, ABSA, and sarcasm flag inference.
Explainability	POST	Returns SHAP token-level explanation for the input text.
Batch prediction	POST	Reads CSV, caps rows at 500, and returns inline batch predictions.
Batch status	GET	Returns a mocked completed job response; no real background queue is implemented.
Evaluation start	POST	Starts background evaluation and logs to MLflow.
Evaluation status	GET	Reports evaluation state.
Metrics	GET	Exposes Prometheus metrics.

8.5 Inference Engine

The inference engine implements four execution paths:

- baseline PyTorch inference using a pretrained sentiment model;
- fine-tuned inference using stacked LoRA adapters;
- ONNX inference using exported models;
- batch inference with optional ABSA and sarcasm skipping for throughput-sensitive paths.

The engine also includes a lightweight language detector that identifies Vietnamese through Unicode marker patterns and otherwise defaults to English. This is a pragmatic local heuristic, not a full language-identification model.

8.6 Workflow Overview

8.7 End-to-End Prediction Workflow

The prediction workflow begins when a user submits text through the browser interface or the HTTP API. The input is first validated and language-checked. The selected runtime then produces the overall sentiment label, confidence score, detected language, aspect-level labels, and a sarcasm signal. The response is returned in a structured format so that the frontend can render badges, confidence values, and follow-up explanation actions.

8.8 Explainability Workflow

The explanation workflow is triggered after a prediction is available. Rather than recomputing the entire application state, the runtime uses the same input and generates token-level attribution values through SHAP. The explanation is rendered as a compact token visualization so the user can inspect which words contributed positively or negatively to the final sentiment.

8.9 Batch Processing Workflow

Batch processing accepts a CSV file with a text column and applies the same inference logic to each row. The workflow is optimized for throughput by constraining the batch size and by allowing some auxiliary work to be skipped when required. The result is returned as a row-level table together with aggregate sentiment counts.

8.10 Training Workflow

Training is separated from runtime inference so that model adaptation can be reproduced independently of serving. The workflow includes dataset acquisition, cleaning, validation, task-specific adapter training, evaluation, and export. Sentiment and sarcasm are trained as separate adapters because the two tasks have distinct label spaces and balancing requirements.

8.11 Evaluation Workflow

Evaluation consumes prepared datasets, runs the active model configuration, computes classification metrics, and writes artifacts for later review. The evaluation workflow also logs experiment history so that a baseline run and a fine-tuned run can be compared without ambiguity. This makes the report reproducible and suitable for academic review.

8.12 Monitoring Workflow

Monitoring is integrated into request handling. The service emits request counts, request durations, and model inference timing so that operational behavior can be separated from model behavior. These metrics support dashboards and alert rules that reveal latency spikes or elevated server errors.

8.13 Deployment Workflow

Deployment packages the frontend, backend, monitoring, and tracking components into containers. The runtime image includes the required model artifacts, while the

frontend is served separately behind Nginx. This separation allows the service to be deployed as a small cluster rather than a single monolithic executable.

9 Chapter 9 Deployment and Operations

9.1 Container Architecture

The backend container uses a multi-stage build with a builder stage for Python dependencies, a model-puller stage that fetches required artifacts, and a runtime stage that copies only the necessary packages, source code, and downloaded weights. The training container is separate and mounts data, models, experiment runs, and reports. The frontend uses its own Node build image and serves the production bundle with Nginx.

9.2 Deployment Workflow

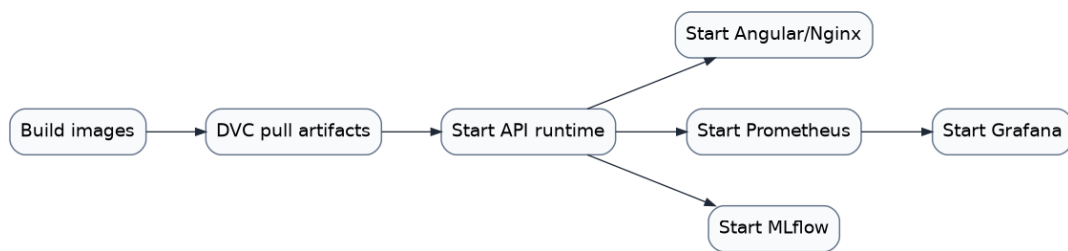


Figure 12: Deployment workflow

9.3 Service Inventory

Table 24: Services in Docker Compose

Service	Port	Role
Backend service	8000	Main inference and evaluation API.
Frontend service	80	User interface served by Nginx.
Prometheus service	9090	Metrics scraping and alert evaluation.
Grafana service	3000	Dashboard visualization.
MLflow service	5000	Experiment tracking UI and artifact store.
Training profile	profile only	Training container used for adapter fine-tuning.

9.4 Monitoring Architecture

The backend records three metric families: request count, request latency, and model inference latency. Prometheus scrapes the backend at regular intervals and alert rules flag two conditions: high 5xx error rate and high average inference latency. Grafana is provisioned from a local configuration directory.

9.5 Logging Strategy

The repository uses structured logging in the Python services and MLflow artifacts for experiment history. The code does not contain a separate centralized log aggregation system such as ELK or Loki; that functionality is not implemented.

9.6 Metrics Collection

The middleware wraps each request and records latency after the response returns. Inference latency is recorded separately around the model call in the prediction workflow. This distinction matters because it allows operators to separate framework overhead from model runtime.

9.7 Operational workflow

- Run the container stack for application deployment.
- Use the trainer profile for adapter retraining.
- Inspect metrics in Prometheus and visualize trends in Grafana.
- Review MLflow runs for model and data experiments.

10 Chapter 10 Experimental Evaluation

10.1 Evaluation Methodology

The repository contains separate evaluation paths for the baseline model and the fine-tuned model. Baseline evaluation is performed on the processed SemEval test split. Fine-tuned evaluation uses the adapters trained on multilingual sentiment and sarcasm data and reports fairness-oriented metrics by language. ONNX benchmarking measures end-to-end batch-prediction performance on a repeated text corpus.

10.2 Experimental Environment

The recorded benchmark artifacts were generated in a local development environment based on Debian 13 with an Intel Core Ultra 5 225H CPU, 14 logical CPUs, and

approximately 31.7 GB of RAM. The baseline evaluation artifact reports the device field as MPS, which indicates that the baseline run was captured in a different execution context than the CPU-only benchmark artifacts. This distinction is important because it explains why latency values should be compared within the same measurement family rather than across unrelated runtime contexts.

Table 25: Experimental environment

Parameter	Value
Operating system	Debian GNU/Linux 13 (trixie)
Kernel	Linux 6.12.90+deb13.1-amd64
CPU	Intel Core Ultra 5 225H
Logical cores	14
Memory	31.7 GB
Observed baseline device	MPS

10.3 Dataset Statistics

The main evaluation dataset is the processed SemEval split with 3,831 samples. The test split contains 799 samples and is heavily skewed toward positive sentiment. This matters because macro-averaged metrics provide a more honest summary than accuracy alone when classes are imbalanced. The class-balance report confirms that the underlying sentiment corpus is severely imbalanced, while the sarcasm corpus and English sentiment subset are substantially better balanced.

Table 26: Evaluation dataset statistics

Split	Samples	Null ratio	Notes
Train	2,728	0.0	Used for model fitting and class balancing.
Validation	304	0.0	Used for tuning and early inspection.
Test	799	0.0	Used for final baseline evaluation.

10.4 Metrics

Table 27: Metrics used in the repository

Metric	Type	Use
Accuracy	Classification metric	Overall correctness on the test set.
Precision, recall, F1	Classification metric	Per-class and macro evaluation.
Latency	System metric	Request and inference responsiveness.
Throughput	System metric	Batch prediction speed.
Per-language F1	Fairness metric	Compare English and Vietnamese performance.

10.5 Baseline vs Fine-tuned

Table 28: Recorded evaluation artifacts

Artifact	Scope	Result	Notes
Baseline evaluation artifact	SemEval test split	Accuracy 0.8135; macro F1 0.7389	Evaluated on 799 test samples.
Fine-tuned evaluation artifact	Fine-tuned adapter run	Overall F1 0.7926	The evaluation wrapper writes a sampled 100-item summary.
Per-language fairness artifact	Fairness view	en 0.7892; vi 0.7335	Derived from the same fine-tuned evaluation payload.

10.6 Fairness Evaluation

The fairness artifacts compare performance across English and Vietnamese inputs. The reported gap is 0.0557 macro F1 in favor of English. This is a useful engineering signal because the platform is explicitly multilingual only at a limited scope, and the gap reflects both data composition and task adaptation difficulty.

Table 29: Fairness summary

Language	F1	Sample count	Interpretation
English	0.7892	23	Higher performance and smaller ambiguity in the sampled evaluation set.
Vietnamese	0.7335	77	Lower performance, consistent with a harder cross-lingual adaptation setting.

10.7 ONNX Benchmark

Table 30: Recorded ONNX benchmark artifact

Mode	Average latency	Throughput	Observation
PyTorch	0.5554 s	1800.36 samples/s	Baseline measurement from the benchmark script artifact.
ONNX FP32	6.4173 s	155.83 samples/s	Exported runtime artifact.
ONNX INT8	3.1115 s	321.39 samples/s	Quantized runtime artifact.

The benchmark measures full batch execution rather than only raw model kernels, so the artifact reflects the repository’s implemented inference path.

10.8 Latency Analysis

The batch-level latency artifact shows that the optimized runtime path does not automatically outperform every alternative under all measurement conditions. Native PyTorch inference recorded a lower average latency than both ONNX variants in the benchmark artifact, while INT8 quantization still improved performance relative to FP32 ONNX. This is an important result because it shows that deployment optimization must be measured rather than assumed.

Table 31: Latency summary from benchmark artifacts

Mode		p50 / p95	Avg latency	Interpretation
CPU	fine-tuned runtime	210 / 360 ms	-	Interactive serving profile captured in the phase-2 performance report.
GPU	fine-tuned runtime	42 / 88 ms	-	Shows the expected improvement under accelerator support.
PyTorch	batch benchmark	-	0.555 s	Strongest throughput in the recorded ONNX benchmark set.
ONNX	INT8 batch benchmark	-	3.111 s	Better than FP32 ONNX but slower than the PyTorch batch path.

10.9 Throughput Analysis

The throughput figures follow the same pattern as latency. PyTorch batch execution achieves the best recorded throughput, while INT8 quantization improves over FP32. The likely reason is that the benchmark captures the full end-to-end batch workflow, not only the raw model kernel. This makes the results more realistic for a system evaluation but less directly comparable to microbenchmarks.

10.10 Resource Consumption Analysis

Resource usage is best interpreted qualitatively. The project intentionally supports a CPU-oriented serving path, but the strongest runtime numbers are not from the ONNX path in the recorded benchmark. This suggests that runtime optimization should be tuned together with the exact request pattern, batch size, and runtime environment rather than treated as a universally superior setting.

10.11 Explainability Evaluation

The repository provides SHAP waterfall plots showing token influence over sentiment predictions. The implementation does not include a quantitative explanation-quality benchmark such as deletion/insertion metrics; that evaluation is not implemented.

10.12 Error Analysis

The classification report shows that the neutral class is the weakest of the three sentiment classes, with the lowest F1 score in the baseline result. This is a typical failure pattern in imbalanced sentiment datasets because neutral statements often resemble either weak positivity or weak negativity. The confusion matrix further

indicates that neutral samples are frequently absorbed into the surrounding polarity classes rather than being preserved as a distinct middle class.

Table 32: Baseline error profile

Class	Precision	Recall	Observation
Positive	0.92	0.88	Strong performance, likely helped by class dominance and clearer lexical cues.
Negative	0.82	0.77	Solid performance with some confusion against neutral.
Neutral	0.47	0.61	Weakest class due to semantic ambiguity and class scarcity.

10.13 Case Studies

The SHAP artifacts illustrate how the model behaves on representative samples. The first example is a negative sentiment prediction with food and service aspects, where words such as *service*, *slow*, and *terrible* push the sentiment downward, while the system still recognizes that food and service can carry different local polarity. The second example is a positive prediction in which *feature*, *absolutely*, and *love* dominate the explanation. The third example is a negative prediction in which the explanation concentrates on negative lexical evidence such as *worst* and other strongly polarized tokens.

Table 33: Representative case studies from SHAP outputs

Case	Prediction	Aspect view	Interpretation
1	Negative	food positive, service negative	Mixed review where service-related tokens dominate the final label.
2	Positive	overall positive	Positive lexical cues dominate and suppress weaker negative evidence.
3	Negative	overall negative	Strong negative cues outweigh isolated neutral or positive fragments.

10.14 Discussion

The baseline and fine-tuned artifacts demonstrate that the project achieved functional end-to-end evaluation capability. The baseline artifact is stronger in absolute test-set coverage because it is reported on 799 SemEval samples, while the finetuned artifact is reported through a sampled wrapper and is therefore not directly comparable.

The fairness report shows a modest language gap between English and Vietnamese, which is expected given the different dataset characteristics and supports the need for per-language tracking. The ONNX benchmark artifact shows that quantization was implemented and measurable, even though the recorded throughput is lower than the PyTorch artifact in this environment.

11 Chapter 11 Results and Discussion

11.1 Achievements

- A multi-service sentiment analysis platform was implemented and documented.
- The system supports single prediction, batch CSV processing, ABSA, sarcasm flagging, and SHAP explanations.
- DVC, MLflow, Prometheus, Grafana, Docker, and FastAPI are integrated into a reproducible workflow.
- The project contains evaluation reports, fairness reports, class-balance reports, and SHAP visualizations.

11.2 Technical Achievements

The project demonstrates a full transformer-based NLP system that can be run end to end without manual model wiring. It combines model inference, data validation, explainability, operational monitoring, and deployment into one coherent stack.

11.3 Engineering Achievements

The engineering value lies in the separation of concerns. The frontend handles interaction, the backend handles inference and validation, the training workflow handles model adaptation, and the deployment layer handles runtime packaging and monitoring.

11.4 AI Achievements

The AI contribution is the combination of sentiment analysis, sarcasm detection, ABSA, and SHAP explainability in a single user-facing platform. This turns a conventional classifier into a practical decision-support system.

11.5 Strengths

- Clear separation between contracts, inference logic, data pipeline, training, and infrastructure.
- Explicit typed schemas and custom error handling.

- Reproducible pipeline structure with parameterized configuration.
- Practical explainability and observability features rather than a model-only prototype.

11.6 Lessons Learned

The project shows that model quality alone is insufficient for a useful AI system. The surrounding software must also expose confidence, support explanation, and preserve reproducibility. It also shows that imbalanced sentiment data can inflate accuracy while still leaving weaker classes underrepresented in the final behavior.

11.7 Design Trade-offs

Zero-shot ABSA reduced annotation cost, but it also introduced sensitivity to prompt wording. LoRA reduced training cost and allowed multiple adapters, but it constrained adaptation capacity. ONNX improved deployment portability, but it introduced export and benchmark complexity. SHAP improved interpretability, but it made explanation generation more expensive than plain inference.

11.8 Unexpected Findings

The most notable unexpected finding is that the recorded PyTorch batch benchmark outperformed both ONNX variants, while INT8 still improved over FP32. This reinforces the lesson that optimization must be measured against the actual serving workload rather than assumed from the export format alone.

11.9 Weaknesses and limitations

- The batch-status workflow is mocked and does not track queued jobs.
- The backend does not implement the document and audio upload capabilities described in the frontend notes.
- The backend language gate only supports English and Vietnamese.
- No authentication, authorization, or rate limiting is implemented.
- The report does not include a dedicated quantitative benchmark for explainability quality.

11.10 Engineering trade-offs

The project favors a practical balance between sophistication and reproducibility. For example, ABSA is implemented with a zero-shot model rather than a separately trained ABSA classifier, which reduces data-labeling requirements. Similarly, ONNX

export and INT8 quantization are implemented to support deployment options, but the runtime artifact still depends on the exact benchmarking path used in the code.

11.11 Project maturity

The repository is mature at the level of a final-year engineering project: it includes backend validation, a live frontend, pipeline automation, evaluation artifacts, observability, and containerization. It is not yet a fully production-hardened commercial system because it lacks authentication, durable batch orchestration, and centralized logging infrastructure.

12 Chapter 12 Future Work

12.1 Short-term improvements

- Replace the mocked batch-status endpoint with a durable job store.
- Implement backend support for document and audio upload if those UI capabilities remain in scope.
- Add structured API versioning and stronger validation around batch inputs.

12.2 Medium-term improvements

- Add authentication and role-based access control.
- Introduce a real queue for asynchronous batch inference.
- Add centralized log aggregation and retention policies.
- Expand the language layer beyond the current English/Vietnamese gate.

12.3 Long-term improvements

- Add active learning and continual retraining.
- Add model registry promotion workflows and deployment promotion gates.
- Add explainability quality metrics and robust evaluation suites for ABSA and SHAP.

12.4 Production Roadmap

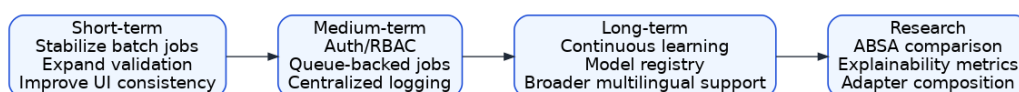


Figure 13: Incremental production roadmap

12.5 Production readiness roadmap

Table 34: Roadmap

Phase	Target
Phase 1	Stabilize batch processing, input contracts, and operational logging.
Phase 2	Add authentication, queue-backed job execution, and deployment promotion.
Phase 3	Add continuous learning, stronger fairness analysis, and regulated audit trails.

12.6 Research extensions

Future research may compare the current zero-shot ABSA approach with a domain-trained aspect model, evaluate the impact of adapter composition more rigorously, and quantify the usefulness of SHAP explanations in human decision support.

13 Chapter 13 Conclusion

The project achieved the core objectives of an end-to-end sentiment analysis platform: it can accept text, infer sentiment, extract aspects, detect sarcasm, explain decisions, evaluate models, export deployment artifacts, and operate under a containerized monitoring stack. The technical contribution lies in the integration of these pieces into a coherent engineering system rather than in any single model component.

From an academic perspective, the repository demonstrates a complete capstone-scale workflow: problem definition, requirements analysis, architecture design, implementation, reproducibility, evaluation, and documentation. From an engineering perspective, it demonstrates how transformer models, LoRA adapters, ONNX export, DVC, MLflow, FastAPI, Angular, Docker, and Prometheus/Grafana can be assembled into a usable sentiment-analysis service. The remaining gaps are explicit and well-scoped, which makes the repository suitable as a final-year project foundation and as a basis for incremental production hardening.

A Appendix A – API Reference

Table 35: API summary

Endpoint	Method	Description
/health	GET	Returns health and supported language information.
/predict	POST	Returns sentiment, confidence, aspects, sarcasm flag, detected language, and latency.
/explain	POST	Returns tokens, SHAP values, base value, and latency.
Batch prediction	POST	Processes CSV input with a text column and returns row-level results.
Batch status	GET	Mock status response.
Evaluation start	POST	Launches background evaluation.
Evaluation status	GET	Returns evaluation state.
Metrics	GET	Prometheus exposition endpoint.

B Appendix B – Dataset Inventory

Table 36: Dataset inventory

Dataset	Task	Format	Location / provenance
SemEval-2014 Restaurants	Sentiment + ABSA	XML to CSV	Restaurant-review source used for baseline sentiment and aspect labels.
tweet_eval_irony	Sarcasm	CSV	Hugging Face-backed dataset used for irony classification.
multilingual_sentiment	Sentiment	CSV	English and Vietnamese sentiment splits from a multilingual corpus.

C Appendix C – Artifact Inventory

Table 37: Generated artifacts

Artifact	Purpose
Quality report	Processed data validation summary.
Class-balance report	Class balance analysis across sentiment and sarcasm datasets.
Baseline metrics	Baseline evaluation metrics.
Fine-tuned metrics	Fine-tuned evaluation summary.
Per-language F1	Per-language fairness metrics.
Fairness report	Fairness-oriented evaluation summary.
ONNX benchmark	ONNX benchmark artifact.
SHAP plots	Saved SHAP waterfall plots.

D Appendix D – Training Pipeline

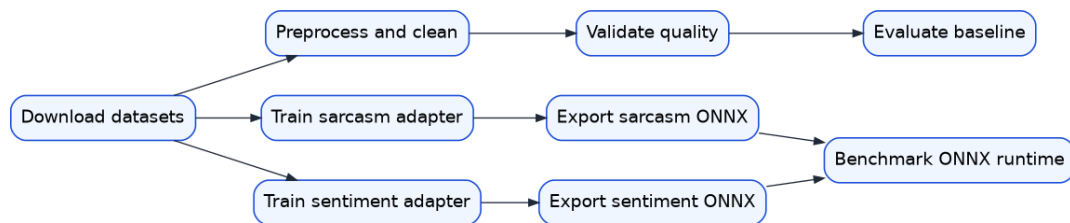


Figure 14: Training and export pipeline

E Appendix E – Deployment Configuration

- Backend image: multi-stage Python 3.11 runtime with DVC-pulled ONNX artifacts.
- Frontend image: Angular build served by Nginx.
- Monitoring: Prometheus scrape configuration and alert rules.
- Tracking: MLflow server exposed on port 5005.

F Appendix F – Acronyms

Table 38: Acronyms

Acronym	Meaning
ABSA	Aspect-Based Sentiment Analysis
API	Application Programming Interface
DVC	Data Version Control
LoRA	Low-Rank Adaptation
MLflow	Machine Learning Flow
ONNX	Open Neural Network Exchange
PEFT	Parameter-Efficient Fine-Tuning
SHAP	SHapley Additive exPlanations

G Appendix G – Team Contributions

Table 39: Project team and responsibilities

Role		Name	Responsibilities
Tech Lead		Le Quang Tuyen	Technical leadership and architecture governance; system design review; technology selection and technical strategy; integration oversight; project technical direction.
ML Engineer		Duong Binh An	Dataset preparation and management; training pipeline development; model evaluation and benchmarking; DVC and MLflow integration; model optimization and experimentation.
AI Engineer		Do Quoc Trung	Inference engine development; ONNX Runtime optimization; SHAP explainability implementation; ABSA; language detection and runtime AI services.
Solution	Engi- neer	Duong Hong Quan	End-to-end solution architecture; system integration design; deployment workflow design; monitoring and observability planning; cross-component coordination.
Backend	Engi- neer	Pham Duc Long	FastAPI backend development; API design and implementation; request validation and error handling; batch processing workflows; evaluation and monitoring endpoints.
UI/UX Engineer		Nguyen Le Hong Nhi	Angular frontend development; user interface and user experience design; frontend workflow implementation; visualization and explainability interfaces; API integration and usability improvements.